FET small-signal I

Cheat sheet · Day 18 · Tue 10 Nov · pack E11

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

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FET SMALL-SIGNAL (Week 8)          ALWAYS: DC Q-point -> gm, rd -> ac circuit
gm0 = 2 IDSS/|VP|        gm = gm0(1 - VGSQ/VP) = gm0*sqrt(IDQ/IDSS)   (JFET, D-MOS)
E-MOS: gm = 2k(VGSQ - VGS(Th))     gm = gfs = yfs      rd = 1/gos = 1/yos
Model: G-S open | gm*Vgs (D->S) || rd | same model for JFET/D-MOS/E-MOS
Device only: Zo = rd, Av(FET) = -gm*rd
CS fixed / self-bias(Cs) / divider(Cs):  Zi = RG or R1||R2,  Zo = RD||rd,
                                         Av = -gm(RD||rd)  (~ -gm RD if rd >= 10RD)
CS unbypassed RS (rd ignored): Zi = RG, Zo ~ RD, Av = -gm RD/(1 + gm RS)
   with rd: Av = -gm RD/(1 + gm RS + (RD+RS)/rd)
Load/source: Av = -gm(rd||RD||RL); Avs = Zi/(Zi+Rsig) * Av; Zo excludes RL
DC reminders: fixed VGS=-VGG; self VGS=-ID RS; divider VGS=VG-ID RS;
   D-MOS with G,S at 0 V -> ID = IDSS, gm = gm0.  Pick root with VGS > VP (or > VGS(T
   Saturation check: VDS > VGS - VP  (E-MOS: VDS > VGS - VGS(Th))
Design for gain: RD||rd = |Av|/gm -> RD = 1/(1/(RD||rd) - 1/rd); recheck VDS
Graphical gm: tangent slope at Q, or chord symmetric about Q (exact for parabola)
Link to BJT: 1/gm plays the role of re.  CS output 180 deg out of phase.
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